



Distributed Information Processing

4th Lecture

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Outline

- Architectures and Algorithms
 - Matching the Architecture to the Algorithm
- Distributed Processing
 - Performance
 - Ability
 - Behavioral Principles
- Q&A

Principal Architectural Models

■ Client-Server

- Client Processes Interacting w/ Server Processes to Access Their Shared Resources Managed in a Centralized Manner

Not Scalable

- E.g., DNS Servers, Web Search Engines

Figure 2.3 on Page 63

Principal Architectural Models (Cont'd)

■ Peer-to-Peer

- Processes Interacting Cooperatively to Access Their Shared Resources Collectively—Managed in a Distributed Fashion

- E.g., Napster

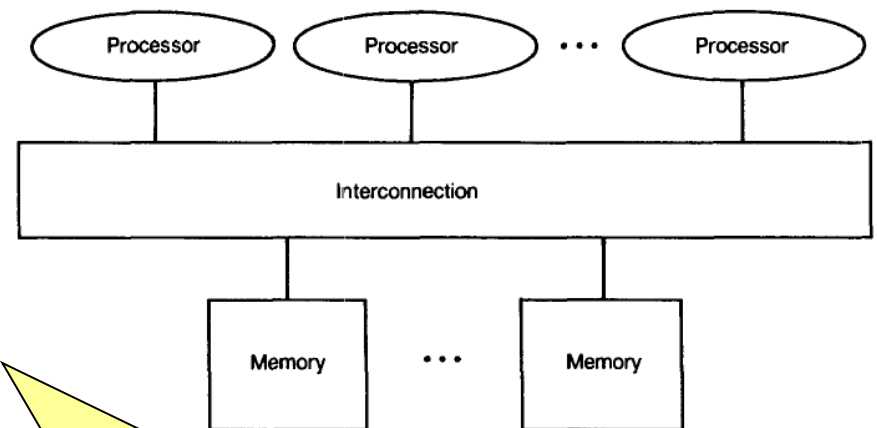
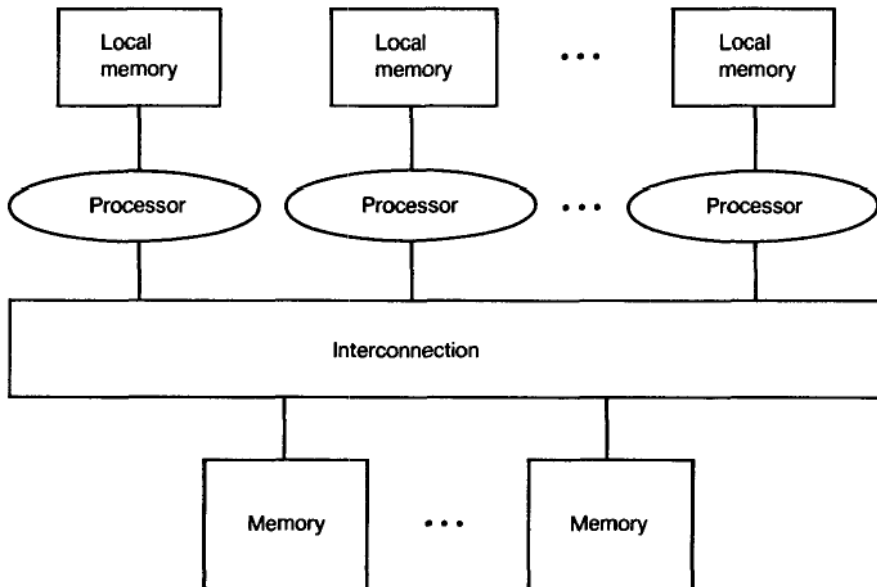
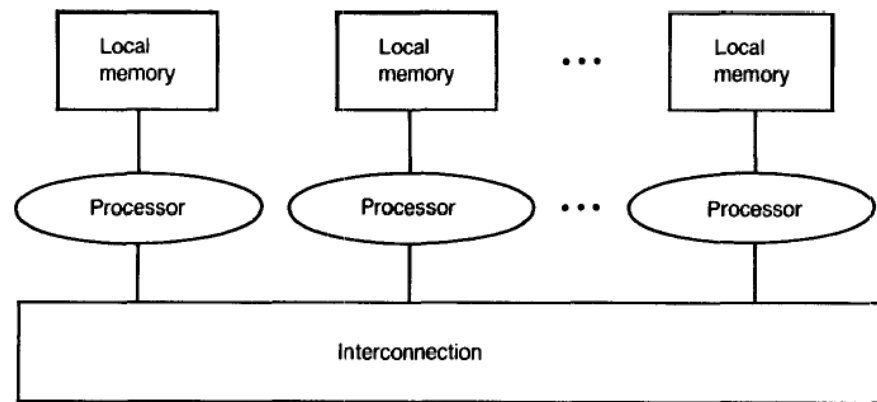
Distributed Load,
Resiliency

Figure 2.4a on Page 48

Flynn's Model for Architectures

- SISD (Single Instruction Stream, Single Data Stream)
 - Uniprocessor
- SIMD (SI, Multiple Data Streams)
 - Single Instruction Memory & Control Processor w/ Memory per Processor
- MISD (Multiple Instruction Streams, SD)
- MIMD
 - Centralized Shared-Memory Architectures
 - E.g., UMA, Uniform Memory Access
 - Distributed-Memory Architectures
 - E.g., NUMA, Non-Uniform Memory Access

MIMD Architectures



E.g., MPP (Massively Parallel Processing)
– Problem: Explicit Data Distribution

E.g., SMP (Symmetric Multiprocessing),
– Problem: Poor Scalability

E.g.,
NUMA

Characterizing the Algorithm [Kleinrock85]

■ To Exploit Its Potential for Concurrency at Different Granularity Levels:

□ Job

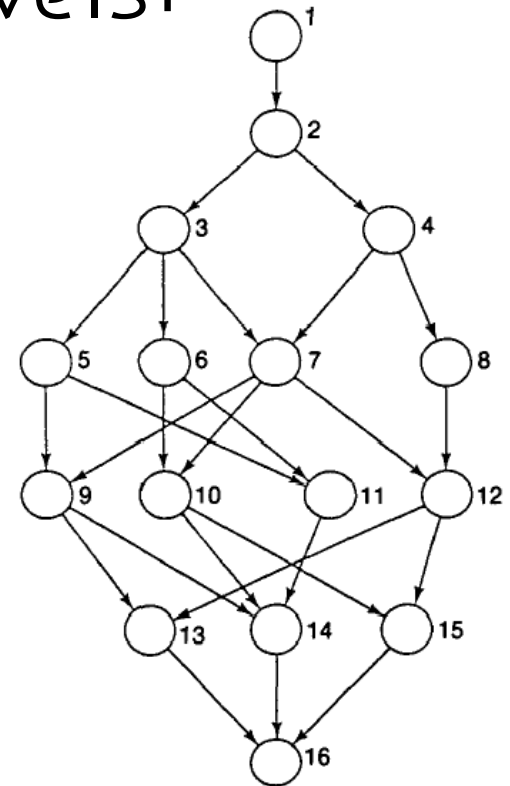
□ Task E.g., by Using a Graph Model

□ Process

□ Instruction

□ Register Transfer

□ Logic Device





Matching the Architecture to the Algorithm [Kleinrock85]

■ Considerations Regarding the Graph Model

□ Partitioning

- Deciding granularity levels
- Grouping

□ Scheduling

- Assigning processors and memory modules

□ Memory Access

□ Interprocessor Communication

□ Synchronization

- Preserving dependency



Matching the Architecture to the Algorithm [Kleinrock85]

- Balance and Tradeoff among Communication, Processing, and Storage
 - Trading Processing for Communication
 - E.g., data compression prior to transmission
 - Trading Storage for Processing
 - E.g., storing a list of computational results
 - Trading Storage for Communication
 - E.g., storing data from a previous communication
- Failure Detection & Recovery

Parallel-Processing Performance

■ Speedup (S)

□ Serial Computation Time T_s over Parallel Execution Time T_p ($S = T_s / T_p$)

Amdahl Portion
– Speedup Will Be
Limited by This Portion

■ $T_p = TCOMP_p + TCOMM_p$

■ $TCOMP_p = T_s(1-A)/P + T_sA$, where A is the serial portion

□ E.g., If $T_s = 100$, $A = 0.2$, & $P = 10$, then $TCOMP_p = 28$

■ $T_p = T_s(1-A)/P + T_sA$, assuming that $TCOMM_p = 0$

■ $S = T_s / T_p = T_s / (TCOMP_p + TCOMM_p) = 1 / \{A + (1-A)/P\}$, assuming that $TCOMM_p = 0$

□ $1 \leq S \leq P$

Ability of Distributed Processing

■ Possibly Producing the Best Result

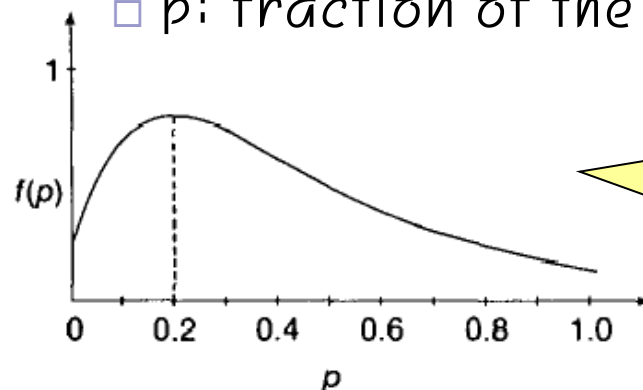
- Greedy Solutions with Unexpected Results:
E.g., Goore Game [Kleinrock85] (Repeating
Actions Preferentially by Trial and Error)

- Many Players Being Asked to Vote YES or NO in
Each Interval

- Acting w/ sufficient memory in a probabilistic fashion
- W/ each unaware of the others

- Referee Rewarding Each Member Independently
w/ a Probability Given by $f(p)$ Unknown to It

- p : fraction of the player set that votes YES



Exactly 20% of the
Players Will Vote YES
with Probability 1

We May Be Able to
Explain How the
Colony of Ants
Perform Its Tasks

Ability of Distributed Processing (Cont'd)

■ Possibly Producing a Globally Suboptimal Solution

□ Most Greedy Solutions: E.g., Prisoner's Dilemma (Predicting the Behavior of the Other Players)

- Two Men Being Held in Separate Cells for a Crime They Did
- District Attorney w/ Hard Evidence Just for One-Year Penalty
 - Making the following offer to each prisoner:
 - Free if confessing w/ your partner remaining silent
 - In jail for 20 years in the opposite case
 - Five years in jail for each if both confess

Dilemma: Each Is Tempted to Confess, But If Both Confess, It Will Be Worse



Behavioral Principles [Kleinrock85]

- Developing innovative architectures for parallel processing
- Providing better languages and algorithms for specification of concurrency
- More expressive models of computation
- Matching the architecture to the algorithm
- Understanding the trade-off among communication, processing, and storage
- Evaluation of the speedup factor for classes of algorithms and architectures
- Evaluation of the cost-effectiveness of distributed-processing networks
- Study of distributed algorithms in networks
- Investigation of how loosely coupled self-organizing automata can demonstrate expedient behavior
- Development of a macroscopic theory of distributed systems
- Understanding how to average over algorithms, architectures, and topologies to provide meaningful measures of system performance